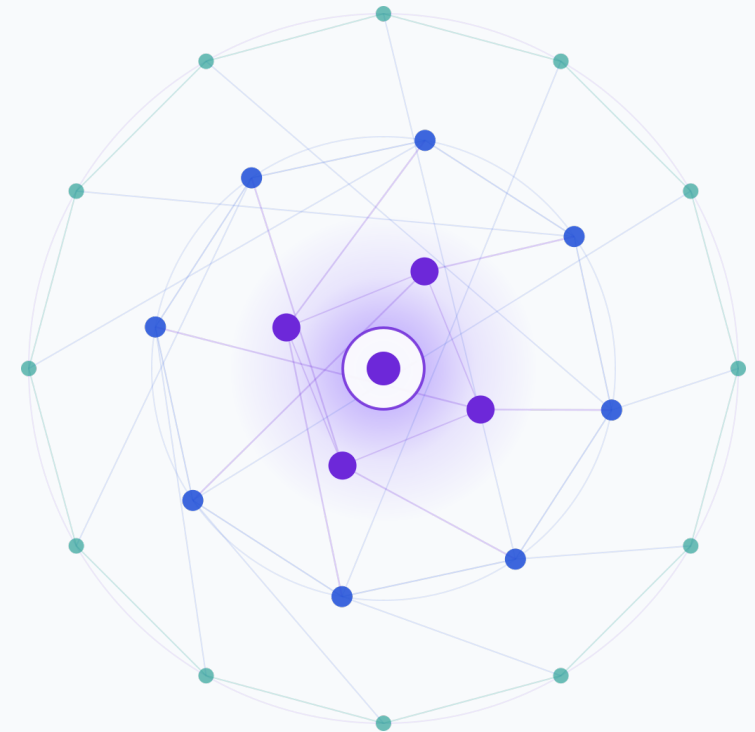


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - Theme 2 — Troubleshooting: *Smart Guided Troubleshooting Engine.*
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- Submission Github link - <https://github.com/KrishnaKapoor612/Nexora-Samsung-Prism.git>



Theme Overview & Strategic Focus

Aligning GenAI capabilities with enterprise support requirements for Samsung Galaxy devices.

CORE PROBLEM STATEMENT

Colloquial User Complaints

Galaxy users describe issues vaguely (e.g., 'my phone is lagging', 'floating bubble won't go away'). Manual triage requires navigating complex, nested Settings menus, leading to high support friction.

ENGINE OBJECTIVE

One-Tap Actionable JSON

Translating non-technical inputs into structured bixby:// deep-link action plans that execute instantly with a single tap, eliminating multi-step manual user navigation.

EXECUTION PRINCIPLES

Zero-Hallucination & Speed

Ensuring strict mathematical post-processing verification, 100% Pydantic schema validation, and sub-15ms response latency via two-stage semantic caching.

Existing Solutions & Industry Pain Points

Current customer support workflows suffer from high latency, hallucinated guidance, and safety risks.

CURRENT SUPPORT LANDSCAPE

- **Manual Agent Triage**

Customer support reps manually search unstructured internal databases (SIIS) and draft manual step-by-step text responses.

- **Generic LLM Chatbots**

Standard conversational AI models generate generic text advice without direct integration into device Settings or deep-link navigation.

THE 4 CRITICAL GAPS

- ✗ **High Latency Triage**

Manual handling takes ~15 minutes per complaint scenario globally.

- ✗ **Hallucinations & Broken Links**

Standard LLMs frequently invent non-existent web URLs and dead Settings paths.

- ✗ **Navigation Friction**

Non-technical users struggle to manually locate complex nested Galaxy Settings menus.

- ✗ **Destructive Action Safety**

Existing systems lack mechanisms to sort risky actions (e.g., Factory Reset) safely to the bottom.

Our Solution: Enterprise RAG & Post-Processing

A multi-layered AI RAG pipeline engineered specifically for high-accuracy device troubleshooting.

01

Hybrid Evidence Retrieval

Combines BM25 keyword matching with Dense Vector Embeddings over local SIIS knowledge bases and live samsung.com/support web scraping for grounded context.

02

LLM + Mathematical Processing

Leverages Amazon Nova-Lite for context understanding, enforced by programmatic constraints (5-7 word limits, atomic step splitting, and strict safety sorting).

03

Intent Deeplink Mapping

Dynamically maps extracted troubleshooting steps to native `bixby://` URIs using Jaccard similarity scoring and navigation vocabulary bonuses.

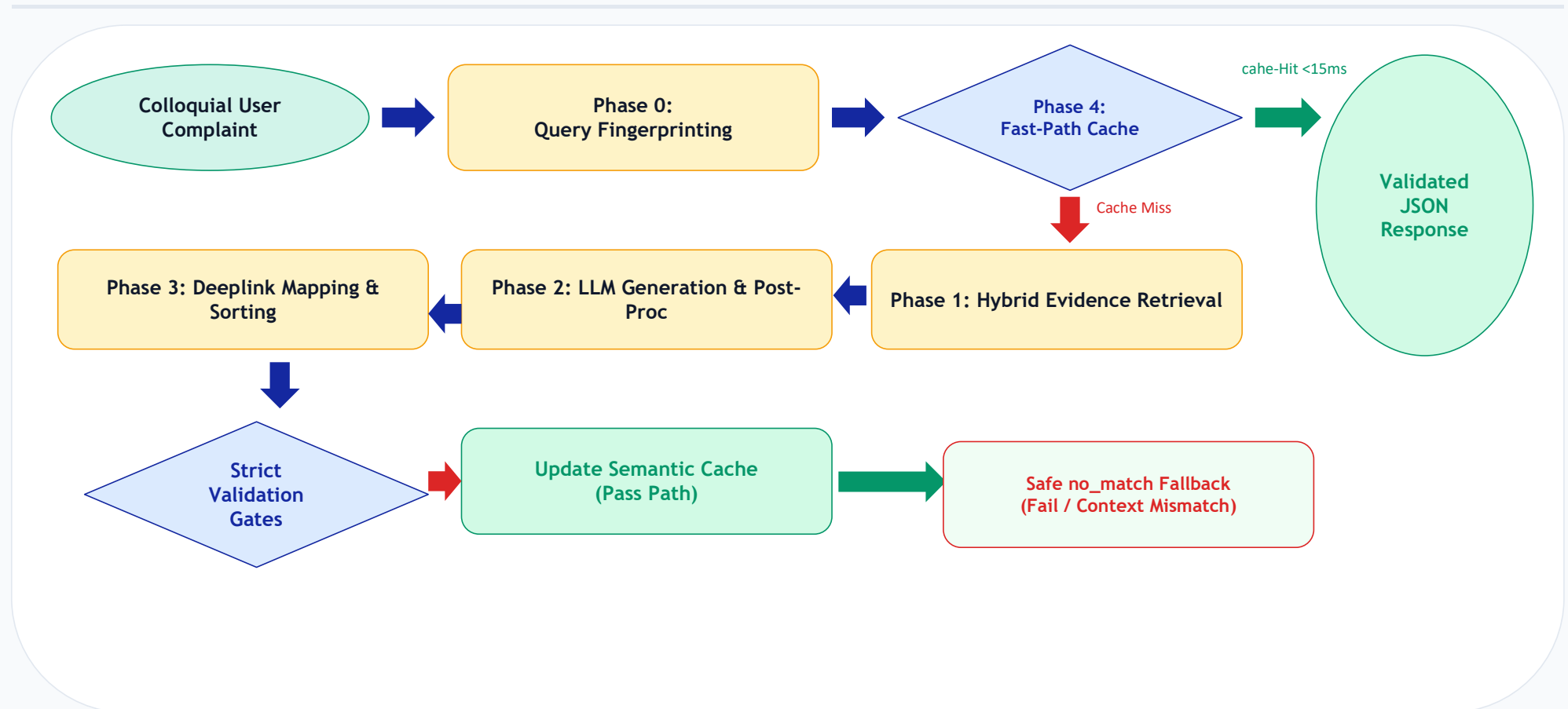
04

Fast-Path Semantic Cache

Captures user query intent and serves paraphrased complaints in < 15ms, completely bypassing LLM inference for cached issues.

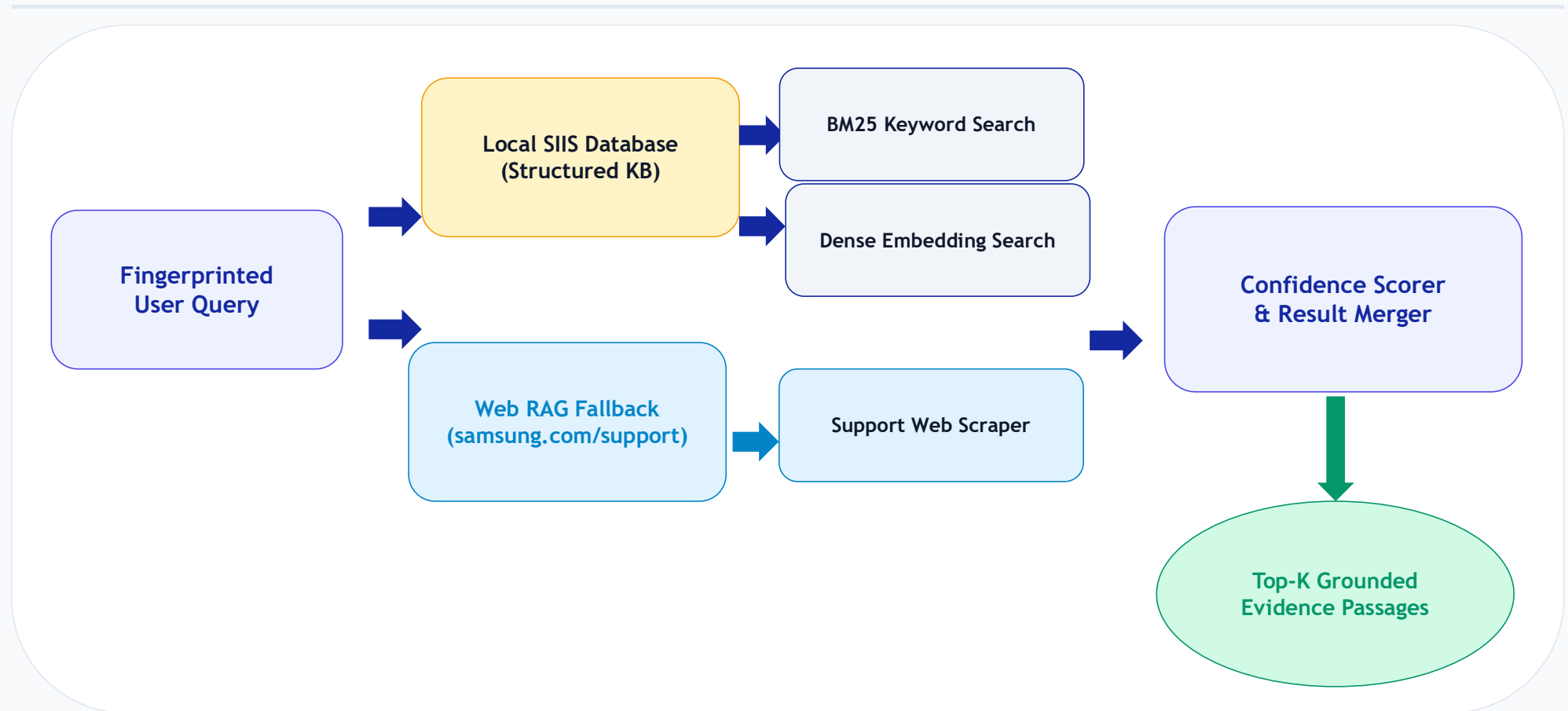
System Architecture: End-to-End Workflow

Diagram 1: Sequential processing from colloquial input to validated JSON response.



Phase 1: Hybrid Evidence Retrieval Architecture

Diagram 2: Dual-engine retrieval over local SIIS database and live Web RAG scraper.



Demo & Product Walkthrough Steps

Validating system performance across local query execution, caching, and fallback handling.

video-demo-link-given-below :

https://drive.google.com/file/d/11TPDGdQT_lbf46NN1c_FHGc-hYI35ISP/view?usp=sharing

Tools & Technology Stack

Modular enterprise stack built for performance, security, and effortless deployment.

BACKEND & API LAYER

FastAPI, Pydantic, Python 3.11

High-throughput asynchronous web framework with strict schema validation.

AI & RETRIEVAL ENGINE

Amazon Nova-Lite, Sentence-Transformers, BM25

Hybrid RAG pipeline combining dense embedding vectors with keyword search.

FRONTEND INTERFACE

Streamlit (Custom Samsung One UI CSS)

Responsive, mobile-first frontend mimicking native One UI design guidelines.

DEPLOYMENT & INFRA

Docker Containerization

Clean, reproducible containerized packaging for seamless enterprise deployment.

Measurable Impact & Enterprise Value

Quantifiable efficiency gains across resolution speed, user friction, and system safety.

15 Min → < 8 Sec

Massive Time Savings

Reduces traditional support triage from ~15 minutes down to < 8 seconds (or < 15ms for cached queries).

1-Tap Deep Links

Frictionless User Experience

Translates colloquial inputs ('phone is lagging') into direct bixby:// setting action buttons.

Zero Destructive Risks

Enterprise Safety Safeguards

Prevents accidental or out-of-order execution of destructive actions like Factory Reset.

Innovation Highlights, Results & Limitations

Transparent assessment of technical achievements and identified system boundaries.

INNOVATION HIGHLIGHTS

- **Mathematical Post-Processing**

Enforces strict step length (5-7 words) and atomic step splitting.

- **Two-Stage Semantic Cache**

Delivers sub-15ms response latency for paraphrased queries.

VALIDATED RESULTS

- **100% Schema Compliance**

Strict Pydantic models guarantee valid JSON outputs.

- **0 Hallucinated URLs**

Programmatic verification eliminates non-existent web links.

KNOWN LIMITATIONS

- **Web RAG Fallback Scraper**

Without official Samsung Search API access, scraper occasionally pulls from marketing pages rather than support articles.

Future Roadmap: Strategic Next Steps

Expanding capability horizons through multimodal AI vision and real-time device state telemetry.

PHASE I

Multimodal Vision Diagnostics

Allowing users to capture screenshots of broken UI or error codes. Vision models will extract error states and map them directly to native `bixby://` deep links.

PHASE II

Real-Time Device State Telemetry

Integrating directly with on-device hardware diagnostics (e.g., automatically pulling battery cycle count and health metrics when a user reports battery drain).

Competitive Differentiating Factors

Key architectural advantages that elevate our solution beyond standard RAG wrappers.

01

Zero-Hallucination Guardrails

Rather than relying solely on prompt engineering, our architecture implements an active mathematical validation layer that verifies generated steps against scraped evidence.

If the LLM generates unsupported steps, the validator catches the token mismatch and securely degrades to a safe fallback.

02

Production Containerization

Fully packaged with Docker for instant, reproducible, and cloud-native deployment across any infrastructure.

Guarantees consistent execution environment across local development, testing, and production servers.

Public Deliverables & Submission Checklist

Verifying completion status across repository, documentation, demo, and presentation assets.

✓ **Working Prototype Code Base — [COMPLETED (Y)]**

Public GitHub Repository with complete backend, frontend, and RAG pipeline code.

✓ **Reproducible README Documentation — [COMPLETED (Y)]**

Detailed setup, environment config, and Docker execution instructions.

✓ **Product Demo Video — [COMPLETED]**

Max 5-minute walkthrough video demonstrating live query execution & fallback.

✓ **Presentation File — [COMPLETED (Y)]**

Fully structured, editable presentation deck (PPT / PDF).

Thank You

Smart Guided Troubleshooting Engine | Team Nexora

Organised by the Language AI Team and the PRISM Team, Samsung R&D Institute India